

APPLICATIONS

BAYESIAN ESTIMATION OF GROUND-WATER LEVELS

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Summary: A Bayesian estimation procedure is suggested as an alternative to the well-known kriging method in estimating ground-water levels. The Bayesian method uses what Omre (1987) called "qualified guesses" as prior information. The knowledge of the topography is used to determine qualified guesses in this study and a comparison of these Bayes estimates with various kriging estimates shows much smaller errors. The procedure proposed here differs from that given by Omre (1987) in that no weights have to be calculated through minimization of the variance of the estimator.

1. INTRODUCTION

The ground-water levels of 57 boreholes in the Dewetsdorp aquifer have been used to estimate the ground-water levels of 15 other holes in the same area as shown in figure 1. Since the ordinary kriging method, named for Krige (1951) with extensions developed in Journel and Huijbregts (1978), Matheron (1973) and Van Tonder and Botha (1984), did not give good results in this study, the Bayesian method of estimating the water levels has been considered. As in the work by Omre (1987), qualified guesses are introduced as prior information on the water levels. The method introduced by Omre is a Bayesian kriging method in that the unconditional variance of the estimator which is expressed as a linear function of the sample values, is minimized. Omre and Halvorsen (1989) show that Bayesian kriging, as defined by Omre (1987), consists of a family of models of which simple and

universal kriging are both members. The method introduced here is not a kriging method in the sense that coefficients of a linear function have to be obtained such that the variance is minimized, but is a pure Bayesian estimation procedure. In a sense the formulae is similar to that of kriging with external drift (see Ahmed and de Marsily (1987)), where a covariate is incorporated in the trend function in much the same way as the Bayesian approach incorporates the qualified guess.

2. THE BAYESIAN FRAMEWORK

Let $Z(x)$ denote the regionalized variable associated with the site $x \in D$, where D is the domain. Also

$$\underline{Z}(X) = (Z(x_1), \dots, Z(x_n))'$$

is the $(n \times 1)$ column vector of all the observations.

We assume that $Z(x_i)$ is normally distributed with

$$(2.1) \quad E[Z(x_i) | M(x_i) = m(x_i)] = m(x_i), \quad x_i \in D,$$

and

$$(2.2) \quad \text{Var}[Z(x_i) | M(x_i) = m(x_i)] = \sigma^2 \quad \forall x_i \in D,$$

$$(2.3) \quad \text{Cov}[Z(x_i), Z(x_j) | m(x_i), m(x_j)] = \sigma^2 \alpha \rho^{h_{ij}},$$

where $0 < \alpha \leq 1$, $0 < \rho < 1$, and $h_{ij} \geq 0$ is a monotone increasing function of only $d(x_i, x_j)$, the distance between sites x_i and x_j with $i \neq j$. The assumptions are therefore that

- (i) the variances at all the locations in the domain of x are equal and
- (ii) the correlation between observations at different sites is positive and a decreasing function of the distance between the sites.

The assumption of a power decaying spatial correlation in (2.3) seems reasonable in a variety of applications, not only the present one. A similar spatial correlation function was used by Gelhard et al. (1977), Sisson and Wierenga (1981) and Hoeksema and Kitanidis (1985), who define the

covariance function as

$$(2.4) \quad \text{Cov}[z(x_i), z(x_j) | m(x_i), m(x_j)] = V_n \delta_{ij} + V_s \exp\left\{-\frac{h_{ij}}{\ell}\right\},$$

where $h_{ij} = d(x_i, x_j)$ and ℓ is called the integral scale, with δ_{ij} Kronecker's delta. The parameters V_n and V_s measure the "unstructured" and "structured" property variability. In our notation it follows that

$$\exp\left[-\frac{1}{\ell}\right] = \rho, \quad V_s = \alpha\sigma^2 \quad \text{and} \quad V_n = (1 - \alpha)\sigma^2,$$

so that α can be considered the proportion of total variability attributable to spatial separation (Hoeksema and Kitanidis (1985)). This means that conditional on the $m(x_i)$'s, $Z(x)$ is second-ordered stationary around its expectation.

$$(2.5) \quad \text{Let} \quad \underline{M}(X) = (M(x_1), \dots, M(x_n))'$$

with prior distribution

$$\underline{M}(X) \sim N(\underline{\mu}(X), k\sigma^2 T)$$

where

$$T = [t_{ij}]; \quad n \times n$$

and

$$t_{ij} = \begin{cases} 1 & , \quad i = j \\ \alpha\rho h_{ij} & , \quad i \neq j. \end{cases}$$

This means we assume the same covariance structure on $\underline{M}(X)$ as on $\underline{Z}(X)$, except for the scalar factor $k > 0$.

The posterior distribution of $\underline{M}(X)$, given $\underline{Z}(X)$, σ^2 , k , α and ρ is normal with

$$(2.6) \quad E[\underline{M}(X) | \underline{Z}(X)] = (k+1)^{-1} (k\underline{Z}(X) + \underline{\mu}(X))$$

and

$$(2.7) \quad \text{Var}[\underline{M}(X) | \underline{Z}(X)] = k(k+1)^{-1} \sigma^2 T.$$

Suppose now that x_0 is the new site where no observation has been made, and we are interested in the Bayes estimate and posterior distribution of $M(x_0)$, given the observations $\underline{Z}(X)$ at the other n sites. Now, following the covariance structure in (2.5),

$$\underline{M}^*(X) = \begin{pmatrix} \underline{M}(X) \\ M(x_0) \end{pmatrix}$$

is normally distributed with mean

$$\underline{\mu}^*(X) = \begin{pmatrix} \underline{\mu}(X) \\ \mu(x_0) \end{pmatrix}$$

and covariance matrix

$$\Sigma^* = k\sigma^2 \begin{pmatrix} T & t'_0 \\ t_0 & 1 \end{pmatrix},$$

where

$$t_0 = \alpha(\rho^{h_{01}}, \dots, \rho^{h_{0n}}).$$

The distribution of $M(x_0)$ conditional on $\underline{M}(X)$ (and the other parameters) is again normal with

$$(2.8) \quad E[M(x_0) | \underline{M}(X)] = \mu(x_0) + t_0 T^{-1}(\underline{M}(X) - \underline{\mu}(X))$$

and

$$(2.9) \quad \text{Var}[M(x_0) | \underline{M}(X)] = k\sigma^2(1 - t_0 T^{-1} t'_0).$$

The following theorem can now be proved:

Theorem 1.

The posterior distribution of $M(x_0)$ given the data $\underline{Z}(X)$ and the parameters $\underline{\mu}(X)$, m , σ^2 , k , α and ρ , is normal with mean

$$(2.10) \quad \mu^*(x_0) = E[M(x_0) | \underline{Z}(X)] = \mu(x_0) + k(1+k)^{-1} t_0 T^{-1}(\underline{Z}(X) - \underline{\mu}(X))$$

and variance

$$(2.11) \quad \sigma^2(x_0) = \text{Var}[M(x_0) | \underline{Z}(X), \sigma^2] = k\sigma^2(1 + k)^{-1} t_0 T^{-1} t'_0.$$

The proof follows easily from the relationships:

$$E[M(x_0)|Z(X)] = E_{\underline{M}(X)}\{E_{M(x_0)}[M(x_0)|\underline{M}(X)|Z(X)]\}$$

and

$$\begin{aligned}\text{Var}[M(x_0)|Z(X)] &= E_{\underline{M}(X)}[\text{Var}[M(x_0)|\underline{M}(X)|Z(X)] \\ &\quad + \text{Var}[E_{M(x_0)}[\underline{M}(x_0)|\underline{M}(X)|Z(X)]]\end{aligned}$$

If we further assume the invariant non-informative prior, $g(\sigma^2) \propto \sigma^{-2}$, on the variance, theorem 2 follows.

Theorem 2

The posterior distribution of $M(X_0)$, given the data $Z(X)$ and the parameters $\underline{\mu}(X)$, k , α and ρ , is a generalized t -distribution with n degrees of freedom, mean $\mu^*(x_0)$ as in (2.10) and variance

$$(2.12) \quad \text{Var}[M(x_0)|Z(X)] = \frac{k}{k+1} \frac{1}{n-2} (1 - (k+1)^{-1} \underline{t}_0' T^{-1} \underline{t}_0) S$$

where

$$(2.13) \quad S = (Z(X) - \underline{\mu}(X))' T^{-1} (Z(X) - \underline{\mu}(X)).$$

In practice the parameters k , α and ρ would usually be unknown. Appropriate priors on these parameters can solve the problem, but the disadvantages are that

- (i) it will require complicated numerical integration and
- (ii) no obvious non-informative priors exist for fairly general application when there is little prior information concerning these parameters.

Our approach is to determine empirical Bayes estimators for the parameters through the estimated semi-variogram.

The model as given here is related to that of Omre (1987). However, Omre assumes the first two moments of $M(x)$ known, while our assumption is only that the first moment, of "qualified guess", is known. Also, Omre only states that a variogram function and location-invariant spatial correlation must be given. This is done explicitly in the above model, and

the form of the variogram follows directly from the model assumptions.

The main distinction between the two methods, however, is that Omre's kriging technique is an exact interpolation technique, the predicted surface runs through the observations, while this is not the case with the pure Bayesian estimation, the influence of the "qualified guess" will always be there, unless $k \rightarrow \infty$, as seen from (2.5). The reason is that Omre assumes a linear unbiased estimator and the weights are determined to minimize the variance, while the Bayesian estimate is the posterior mean. Kriging itself, however, need not be applied as an exact interpolator, although in practice it usually is.

3. THE SEMI-VARIOGRAM

The variability between two regionalized variables at sites x_i and x_j is characterized by the semi-variogram which is defined as

$$(3.1) \quad \gamma(x_i, x_j) = \frac{1}{2} \text{Var}(Z(x_i) - Z(x_j)).$$

In general, the semi-variogram can be a function of the locations x_i and x_j . Mostly, however, it is assumed that the semi-variogram depends only on the lag, or distance, between the two sites. This is the case for the model in (2.1) - (2.4) for which it follows that the unconditional semi-variogram is given by

$$\gamma(x_i, x_j) = (1 + k)\sigma^2(1 - \alpha\rho^{h_{ij}}),$$

which can be written as a function only of the lag h :

$$(3.2) \quad \begin{aligned} \gamma(h) &= \frac{1}{2} \text{Var}(Z(x) - Z(x+h)) \\ &= (1 + k)\sigma^2(1 - \alpha\rho^h). \end{aligned}$$

When $h \rightarrow \infty$, the semi-variogram reflects the unconditional variance of the observations:

$$(3.3) \quad \gamma(\infty) = (1 + k)\sigma^2,$$

while $\gamma(h)$ should approach the conditional variance of $Z(x)$ when $h \rightarrow 0$.

The assumption is that when $h \rightarrow 0$, two independent observations were taken at the same site, conditional on the mean, both with the same variance, so that

$$(3.4) \quad \gamma(0) = \sigma^2.$$

Unconditionally, two observations from the same site has a covariance $k\sigma^2$, so that (3.4) still holds. The above assumption leads to the relationship

$$(3.5) \quad \alpha = \frac{k}{k+1},$$

so that the semi-variogram can be written as

$$(3.6) \quad \gamma(h) = \sigma^2(1 + k(1 - \rho^h))$$

a function of the two parameters k and ρ .

In geostatistical terms, $\gamma(\infty)$ would be the sill value, while $\gamma(0)$ is the nugget effect. There is a difference between ordinary kriging and the Bayesian analysis in the interpretation of this nugget effect. In kriging $\gamma(0) = 0$ by definition, but there is a discontinuity at the origin which can be caused by measurement errors and other micro-variabilities at an observation site x . The Bayesian view is that there is a natural variability in the measurements at a site x around the real quantity of interest, $m(x)$, so that $\text{Var}[Z(x)|m(x)] = \gamma(0) = \sigma^2$.

For $0 < h < \infty$, an unbiased estimator for $\gamma(h)$ from the marginal distribution of the observations is given by

$$(3.7) \quad \hat{\gamma}(h) = \frac{1}{n(h)} \sum_{h_{ij}=h} [(\alpha(x_i) - \alpha(x_j)) - (\mu(x_i) - \mu(x_j))]^2,$$

where μ is the prior mean, or qualified guesses.

The summation is over the total $n(h)$ of all pairs (x_i, x_j) for which the lag is h . The unbiased estimator for the sill value is

$$(3.8) \quad \hat{\gamma}(\infty) = \frac{1}{n} \sum_{i=1}^n (\alpha(x_i) - \mu(x_i))^2,$$

while an unbiased estimator for $\gamma(0)$ exists only if there is at least one pair of observations from the same site. Then

$$(3.9) \quad \hat{\gamma}(0) = \frac{1}{2n(0)} \sum_{i=1}^n \sum_{\ell=1}^{n_i} (z_{\ell}(x_i) - \bar{z}(x_i))^2,$$

where $n(0)$ is the degrees of freedom; the total number of observations used in the estimation less the number of sites with at least two observations, n_i is the number of observations and $\bar{z}(x_i)$ the sample mean at site x_i .

It is not the purpose of this paper to go into detail of variogram estimation, and the standard estimators are given above. For a discussion of variogram estimation, see Cressie and Hawkins (1980).

The estimator $\hat{\gamma}(h)$ can now be plotted for a sequence of lags and the variogram function (3.6) fitted to these estimates. The resulting function will then yield the estimates for k and ρ .

4. ESTIMATION OF GROUND-WATER LEVELS

A very important application of kriging arises in the estimation of ground-water levels (Delhomme (1978)). Since ground-water levels tend to follow the general topography of the terrain, the data may contain a highly non-linear trend in hilly and coastal areas or in aquifers where large-scale pumping is applied (Gambolati and Volpi (1979)). However, as the topography is usually well known, it could be used, together with other local knowledge, to determine a "qualified guess" for the unknown parameter $M(x)$ in the Bayesian estimation.

The method is applied to the ground-water levels observed in the Dewetsdorp aquifer, where water levels are recorded on a regular basis in 57 boreholes as part of an on-going investigation into the natural recharge of the aquifer. The positions of these boreholes are denoted by "o" in figure 1 and the scale is in surveyor co-ordinates. Fifteen additional boreholes were sunk in the same general area (denoted by "*" in figure 1) and the Bayesian

method is used to estimate the true water levels in the new boreholes and to compare it with the observed water levels.

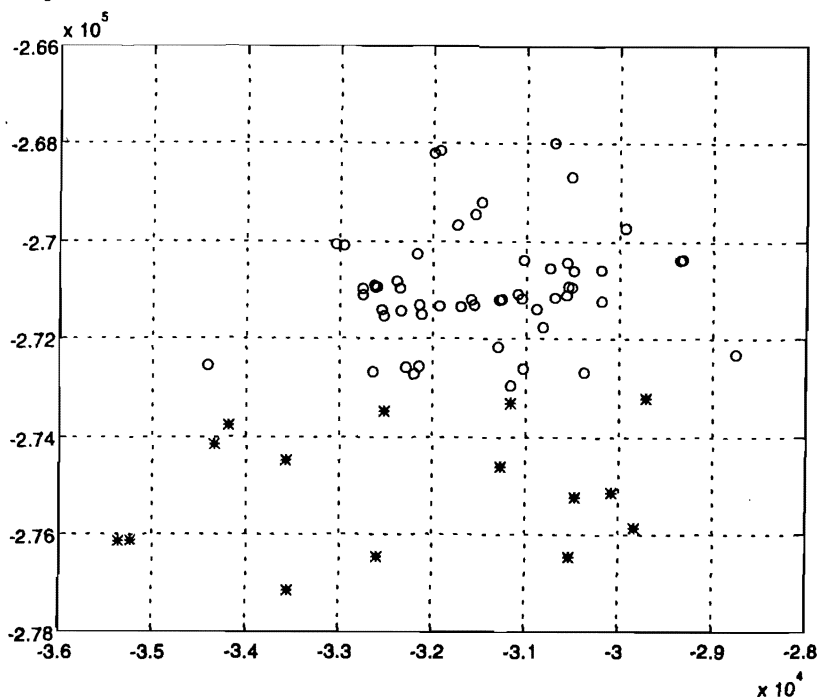


Figure 1. Relative positions of original 57 boreholes (o) and additional 15 boreholes (*) in the Dewetsdorp aquifer.

The distance between two boreholes were used to determine the lag, which is assumed to belong to a finite set of integers. As can be seen from figure 1, some of the holes are very close together, and holes closer than about 40 metres apart were taken to have lag zero. This was taken to be the microscale (Russo and Bresler (1981)), the distance for which the correlation coefficient does not differ significantly from 1. This assumption enables us to obtain an estimate of the semi-variogram at $h = 0$. The lag between holes x_i and x_j were taken as the largest integer smaller or equal to

$0,518d(i, j) - 0,00276d(i, j)^2$, where $d(i, j)$ is the true distance between holes x_i and x_j (in terms of surveyor co-ordinates). This gives a slightly concave function with a maximum lag of 20 among the original 57 boreholes, while the maximum lag among all holes is equal to 24.

The choice of the above formula to determine the lags is subjective and was made in this particular example in order to obtain 20 lags with a reasonable number of observations for estimating the semi-variogram at each lag, and also because an estimate is less sensitive to values at points with a large lag than to points with a small lag. So as the distance between points increases, the interval for each lag increases slightly.

Since this is a very shallow aquifer, the topography of the area was used to obtain qualified guesses for the water heights in the 72 boreholes, and the guesses were taken as linear functions of the collar heights of the holes. The general opinion was that the depth of the aquifer is about 14 metres, depending on the elevation. So the qualified guess was taken as $\mu = \underline{e} - 14 - (\underline{e} - 1570)/100$ metres, where $\underline{e}$ is the elevation of the collar of the borehole and the average height above sea level for the area is 1570 metres. This means a qualified guess of about 14 metres below ground level with a slight correction for topography.

The fitted semi-variogram from the 57 original holes is shown in figure 2 with the estimate for each lag. The point $\hat{\gamma}(\infty) = 12,1$ (from (3.8)) is not shown but was also used in the fitting of the semi-variogram. The Nelder-Mead algorithm (Press et al. (1987)) was used to fit the function (3.6) to the weighted estimates in figure 2.

The estimated parameters are: $\hat{\sigma}^2 = 6,2817$, $\hat{k} = 3,4736$, $\hat{\rho} = 0,6983$.

The last two estimates are used in equations (2.10) and (2.12) to obtain the Bayes estimates and their variances. It is not strictly true anymore that $M(x_0)$ is t -distributed, since the parameters were replaced by estimates, but it is true for fixed k and ρ , and should hold approximately.

In table 1 the estimates and their variances are given for the 15 new boreholes together with the observed water levels. The results obtained by

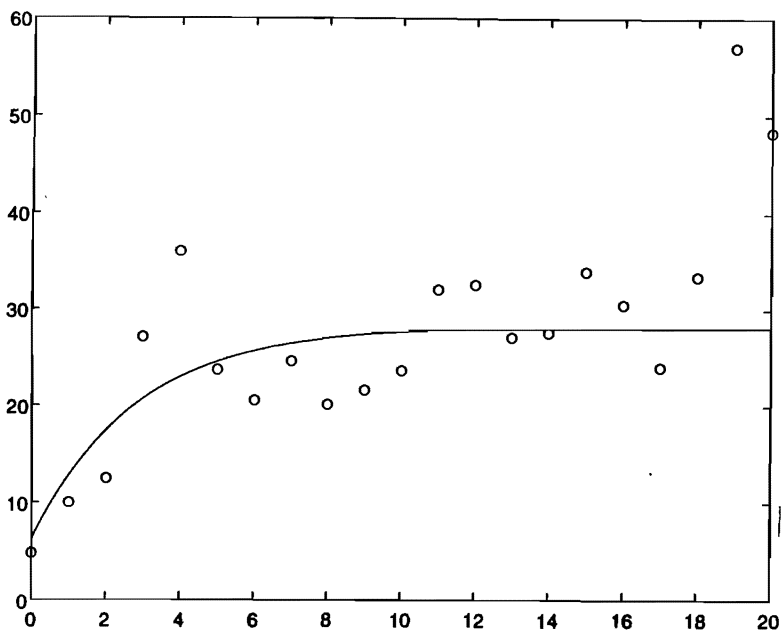


Figure 2. Fitted semi-variogram for 57 boreholes.

applying kriging methods are also given. As can be seen from table 1, ordinary kriging tends to underestimate the water levels. This indicates that the stationarity assumption, inherent in ordinary kriging, does not apply anymore over the area of the new boreholes. This trend is taken account of through universal kriging, which shows a large improvement in mean square error (MSE). The linear model for co-kriging, with collar height as covariate to the water level, performs slightly better than universal kriging, mainly due to the strong correlation between topography and water level. The good performance of the Bayes estimates can also be attributed to good "qualified guesses", based on a function of the collar heights. For co-kriging only the collar heights were used as covariates. In the last column the variances of the Bayes estimates are given from equation (2.12).

TABLE 1. Observed and estimated ground-water levels for 15 holes in the Dewetsdorp aquifer

Borehole	Water levels					
	Observed	Ordinary Kriging	Co-kriging	Universal kriging	Bayes	Variances
1	1603,0	1552,9	1578,53	1614,16	1597,4	33,1
2	1603,6	1552,5	1578,36	1610,98	1597,4	33,1
3	1585,9	1542,4	1559,61	1577,57	1574,4	33,1
4	1610,9	1556,0	1591,07	1579,15	1616,6	33,1
5	1583,0	1557,0	1582,58	1574,99	1603,9	33,0
6	1595,0	1510,5	1566,29	1557,94	1589,4	33,1
7	1571,0	1544,6	1551,73	1562,08	1563,1	33,1
8	1547,0	1502,4	1529,66	1519,60	1545,6	33,1
9	1533,0	1492,4	1517,47	1516,84	1529,4	33,1
10	1518,0	1491,3	1507,12	1521,48	1512,1	33,1
11	1526,0	1503,1	1517,40	1531,50	1527,5	33,1
12	1529,8	1516,0	1537,94	1528,43	1551,2	32,5
13	1493,6	1493,1	1495,08	1496,49	1497,9	32,8
14	1486,2	1499,7	1492,37	1507,56	1492,7	30,8
15	1544,7	1502,6	1526,96	1515,50	1542,3	33,1
MSE		1704,04	311,43	358,21	89,64	

5. CONCLUSION

The example in the previous section is not meant to imply general superiority of the Bayes estimation procedure over kriging. However, the Bayes procedure has the advantage that the introduction of a "qualified guess" enables one to take into account some prior knowledge that can improve the estimate. Such prior information is often not easily quantifiable and thus difficult or impossible to incorporate in a kriging procedure.

For example, it is known that the pump in borehole 5 was switched off shortly before the water-level was measured. This explains the large error in the Bayesian estimate for borehole 5. This information was not used in determining the estimates in table 1, but could easily be introduced by reconsidering the guess for that borehole.

The Bayes estimate for the observation at a site x_0 is a combination of the guess for that site and the observed values at neighbouring sites, and the weight attached to the guess is determined by the reliability of the guesses at sites where observations were made. The value of the constant k determines this weight. A large value of k means that the guesses are reliable and little weight is attached to them. So in the case of no or little prior information, the Bayes estimation is still applicable.

In the example the topography provides good "qualified guesses". However, this is not essential for a successful application of the model. In the case of confined or deep aquifers where there is much more uncertainty about the water levels, topography can still be used to obtain "qualified guesses". If the guesses are poor, it will be reflected by a large estimated value of k . If the user prefers not to use qualified guesses at all, the vector X can be assumed to contain the common element μ , and a locally uniform prior can be placed on μ .

The assumption of constant variance may be restrictive, but is necessary to make the estimation problem identifiable, and should be reasonable if the domain is fairly homogeneous. This restriction can be relaxed if there is more than one observation at all the locations, which would make estimation feasible, or if the variances are assumed known.

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REFERENCES

- ARMED, S., & DE MARSILY, G. (1987). Comparison of geostatistical methods for estimating transmissivity using data on transmissivity and specific capacity. *Water Resources Research*, 23, 1717-1737.
- RESSIE, N. & HAWKINS, D.M. (1980). Robust estimation of the variogram. *Math. Geol.*, 12(2), 125-155.

- DELHOMME, J.P. (1978). Kriging in the hydrosiences. *Advances in Water Resources*, 1(5), 251-266.
- GAMBOLATI, G. & VOLPI, G. (1979). Groundwater mapping in Venice by stochastic interpolators 1 and 2. *Water Resources Research*, 15(2), 281-297.
- GELHARD, L., BAKR, A.A., GUTJAHR, A.L. & MACMILLAN, J.R. (1977). Comments on "A stochastic-conceptual analysis of one-dimensional groundwater flow in nonuniform homogeneous media" by R.A. Freeze. *Water Resources Research*, 13(2), 477-479.
- HOEKSEMA, R.J. & KITANIDIS, P.K. (1985). Comparison of Gaussian conditional mean and kriging estimation in the geostatistical solution of the inverse problem. *Water Resources Research*, 21(6), 825-836.
- JOURNAL, A.G. & HUIJBREGTS, C.J. (1978). *Mining Geostatistics*. Academic Press.
- KRIGE, D.G. (1951). *A statistical approach to some mine valuations and allied problems at the Witwatersrand*. Unpublished M.Sc. thesis, University of the Witwatersrand, Johannesburg.
- MATHERON, G. (1973). The intrinsic random functions and their applications. *Advances in Applied Probability*, No. 5, 439-468.
- OMRE, H. (1987). Bayesian Kriging-merging observation and qualified guesses in kriging. *Mathematical Geology*, 19(1), 25-31.
- OMRE, H. & HALVORSEN, K.B. (1989). The Bayesian bridge between simple and universal kriging. *Mathematical Geology*, 21, 767-786.
- PRESS, W.H., FLANNERY, B.P., TEUKOLSKY, S.A. & VETTERLING, W.T. (1987). *Numerical recipes: The art of scientific computing*. Cambridge University Press.
- RUSSO, E. & BRESLER, E. (1981). Soil hydraulic properties as stochastic processes: I. An analysis of field spatial variability. *Soil Sci. Am. J.*, 45, 682-687.
- SISSON, J.B. & WIERENGA, P.J. (1981). Spatial variability of steady-state infiltration rates as a stochastic process. *Soil Sci. Am. J.*, 45, 699-704.
- VAN TONDER, G.J. & BOTHA, J.F. (1984). *Die toepassing van kriging in geohidrologie*. Bulletin 12, Institute for Ground-water Studies, University of the Orange Free State, Bloemfontein.